SAGAR PUBLIC SCHOOL

Integrated Programme | Topic: Current Electricity – Level 4

PRACTICE SHEET – LEVEL 4

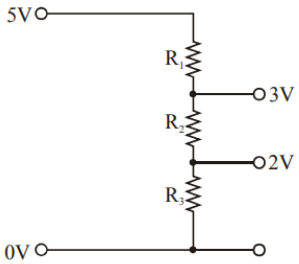
Current Electricity

Time Allowed: 60 Mins **Total Questions: 20**

Name: _____ Class/Sec: _____ Date: _____

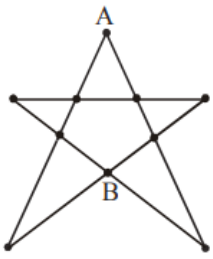
Instructions: Attempt all questions. Each question carries one correct option unless marked *one or more than one option is correct.* Questions without options are numerical-value type: write the answer in the blank. Q.12–14 and Q.17–19 are comprehension sets. A self-assessment page follows; the answer key is provided as a separate sheet.

Q.1 A potential divider is used to give outputs of 2 V and 3 V from a 5 V source, as shown in figure. Which combination of resistances, R_1, R_2 and R_3 gives the correct voltages?



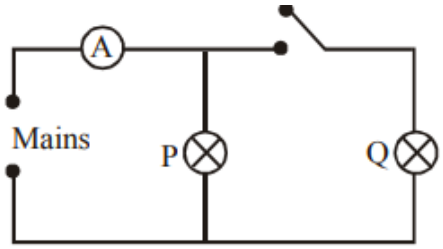
(A) $R_1 = 1\,\text{k}\Omega,\ R_2 = 1\,\text{k}\Omega,\ R_3 = 2\,\text{k}\Omega$
(B) $R_1 = 2\,\text{k}\Omega,\ R_2 = 1\,\text{k}\Omega,\ R_3 = 2\,\text{k}\Omega$
(C) $R_1 = 3\,\text{k}\Omega,\ R_2 = 2\,\text{k}\Omega,\ R_3 = 2\,\text{k}\Omega$
(D) $R_1 = 3\,\text{k}\Omega,\ R_2 = 2\,\text{k}\Omega,\ R_3 = 3\,\text{k}\Omega$

Q.2 The resistance of all the wires between any two adjacent dots is R. The equivalent resistance between A and B as shown in figure is



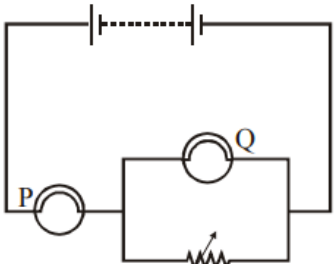
(A) $\frac{7}{3}R$ (B) $\frac{7}{6}R$
(C) $\frac{14}{8}R$ (D) None of these

Q.3 How will the reading in the ammeter A of figure be affected if another identical bulb Q is connected in parallel to P as shown. The voltage in the mains is maintained at a constant value.



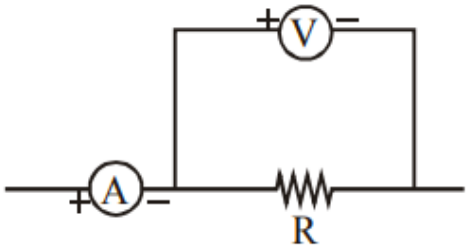
(A) The reading will be reduced to one-half
(B) The reading will not be affected
(C) The reading will be doubled of the previous one
(D) The reading will be increased four-fold

Q.4 The circuit shown in figure, contains a battery, a rheostat and two identical lamps. What will happen to the brightness of the lamps if the resistance of the rheostat is increased?



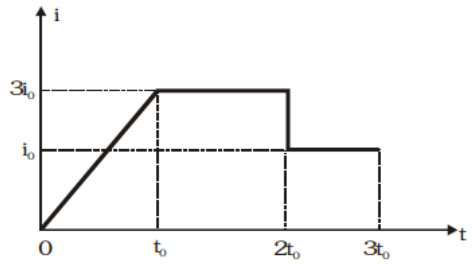
(A) Lamp P: Less brighter; Lamp Q: Brighter
(B) Lamp P: Less brighter; Lamp Q: Less brighter
(C) Lamp P: Brighter; Lamp Q: Less brighter
(D) Lamp P: No change; Lamp Q: Brighter

Q.5 A candidate connects a moving coil voltmeter V, a moving coil ammeter A and a resistance R as shown in figure. If the voltmeter reads 20 V and the ammeter reads 4 A, R is



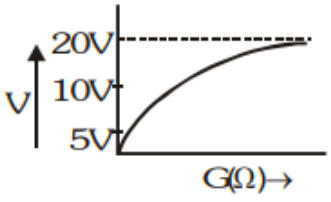
(A) equal to $5\,\Omega$
(B) greater than $5\,\Omega$
(C) less than $5\,\Omega$
(D) greater or less than $5\,\Omega$ depending upon its material

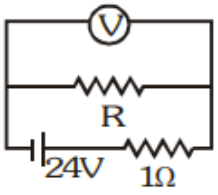
Q.6 A time varying current i is passed through a resistance R as shown in figure. The total heat generated in the resistance is



(A) $11\,i_0^2Rt_0$ (B) $13\,i_0^2Rt_0$
(C) $17\,i_0^2Rt_0$ (D) $15\,i_0^2Rt_0$

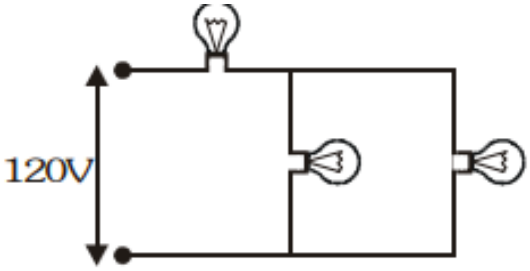
Q.7 A cell of internal resistance $1\,\Omega$ is connected across a resistor. A voltmeter having variable resistance is used to measure potential difference across resistor. The plot of voltmeter reading V against G is shown. What is value of external resistor R? (G = Resistance of galvanometer)





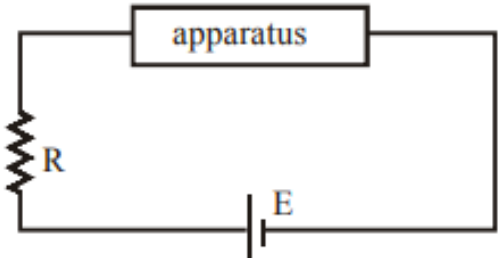
(A) $5\,\Omega$ (B) $4\,\Omega$
(C) $3\,\Omega$ (D) can't be determined

Q.8 Three 60 W, 120 V light bulbs are connected across a 120 V power source. If resistance of each bulb does not change with current then find out total power delivered to the three bulbs.



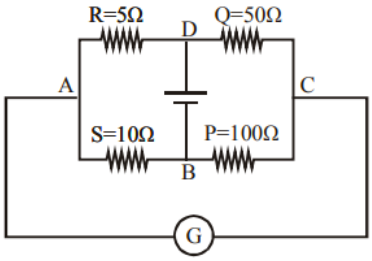
(A) 180 W (B) 20 W
(C) 40 W (D) 60 W

Q.9 An apparatus is connected to an ideal battery as shown in figure. For what value of current, power delivered to the apparatus will be maximum?



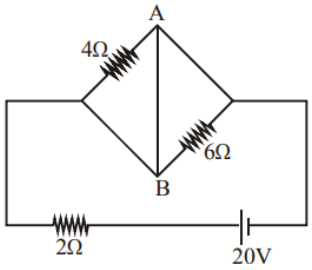
(A) $\frac{E}{R}$ (B) $\frac{E}{2R}$
(C) $\frac{E}{4R}$ (D) information insufficient

Q.10 *(One or more than one option is correct)* Figure shows a balanced Wheatstone's bridge.



(A) If P is slightly increased, the current in the galvanometer flows from C to A
(B) If P is slightly increased, the current in the galvanometer flows from A to C
(C) If Q is slightly increased, the current in the galvanometer flows from C to A
(D) If Q is slightly increased, the current in the galvanometer flows from A to C

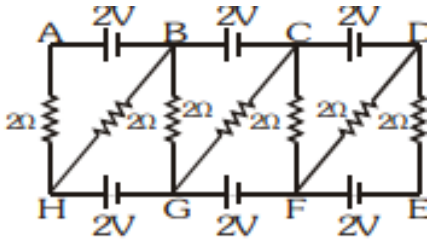
Q.11 *(One or more than one option is correct)* In the circuit shown in figure.



(A) power supplied by the battery is 200 W
(B) current flowing in the circuit is 5 A
(C) potential difference across the 4 Ω resistance is equal to the potential difference across the 6 Ω resistance
(D) current in wire AB is zero.

Q.12 Comprehension (Q.12–14): In given circuit, 7 resistors of resistance 2 Ω each and 6 batteries of 2 V each, are joined together.

The potential difference $V_D - V_E$ is:



(A) $\frac{5}{6}$ V (B) $\frac{6}{5}$ V
(C) $-\frac{5}{6}$ V (D) $-\frac{22}{9}$ V

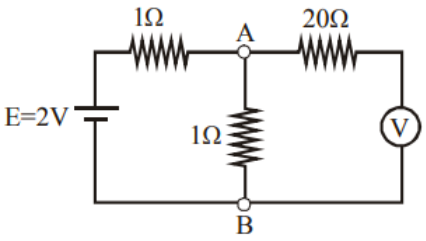
Q.13 (Refer to the circuit of Q.12.) The current through branch BG is:

(A) 1 A (B) 0.2 A (C) 0.4 A (D) 0.6 A

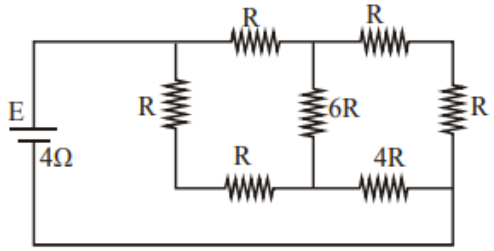
Q.14 (Refer to the circuit of Q.12.) The current through battery between A & B is:

(A) 0.6 A (B) 0.8 A (C) 0.4 A (D) 1 A

Q.15 In the given circuit, the voltmeter and the electric cell are ideal. Find the reading of the voltmeter (in volt).



Q.16 A battery of internal resistance $4\,\Omega$ is connected to the network of resistances as shown in figure. In order that the maximum power can be delivered to the network, the value of R in Ω should be



Q.17 Comprehension (Q.17–19): An ammeter and a voltmeter are connected in series to a battery with an emf of 10 V. When a certain resistance is connected in parallel with the voltmeter, the reading of the voltmeter decreases three times, whereas the reading of the ammeter increases the two times.

Find the voltmeter reading after the connection of the resistance.

(A) 1 V (B) 2 V (C) 3 V (D) 4 V

Q.18 (Refer to the passage of Q.17.) If resistance of the ammeter is $2\,\Omega$, then resistance of the voltmeter is:

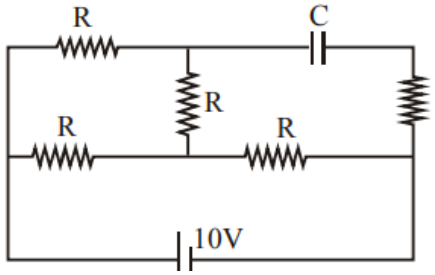
(A) $1\,\Omega$ (B) $2\,\Omega$ (C) $3\,\Omega$ (D) $4\,\Omega$

Q.19 (Refer to the passage of Q.17.) If resistance of the ammeter is $2\,\Omega$, then resistance of the resistor which is added in parallel to the voltmeter is:

(A) $\frac{3}{5}\,\Omega$ (B) $\frac{2}{7}\,\Omega$

(C) $\frac{3}{7}\,\Omega$ (D) None of these

Q.20 Find the potential difference across the capacitor in volts.



SELF-ASSESSMENT – LEVEL 4 SHEET

Time Allowed: 60 Mins	**Time Taken:** _____ Mins
Total Questions: 20	**Questions Attempted:** _____
Correct: _____	**Incorrect:** _____
Unattempted: _____	**Accuracy:** _____ % (Correct / Attempted ×100)

Question-wise record (tick one: C = Correct, W = Wrong, S = Skipped)

Q.1 ☐C ☐W ☐S	**Q.2** ☐C ☐W ☐S	**Q.3** ☐C ☐W ☐S	**Q.4** ☐C ☐W ☐S
Q.5 ☐C ☐W ☐S	**Q.6** ☐C ☐W ☐S	**Q.7** ☐C ☐W ☐S	**Q.8** ☐C ☐W ☐S
Q.9 ☐C ☐W ☐S	**Q.10** ☐C ☐W ☐S	**Q.11** ☐C ☐W ☐S	**Q.12** ☐C ☐W ☐S
Q.13 ☐C ☐W ☐S	**Q.14** ☐C ☐W ☐S	**Q.15** ☐C ☐W ☐S	**Q.16** ☐C ☐W ☐S
Q.17 ☐C ☐W ☐S	**Q.18** ☐C ☐W ☐S	**Q.19** ☐C ☐W ☐S	**Q.20** ☐C ☐W ☐S

Reason for mistakes (tick all that apply)

☐ Concept not clear ☐ Formula forgotten ☐ Calculation mistake
☐ Misread the question ☐ Ran out of time ☐ Guess work

Personal Feedback

What went well:

_____

_____

Topics / questions I need to revise:

_____

_____

My action plan before the next sheet:

_____

_____

Teacher's remarks:

_____

_____

Student's signature: _____ Teacher's signature: _____